

Supplementary Material for “Self-organized Resonance during Search of a Diverse Chemical Space”

Nonequilibrium Generalization of Boltzmann Distribution

There are various ways of deriving Crooks’ microscopic reversibility relation in [7]. Here, for pedagogical purposes, we work in a microcanonical picture. For a Hamiltonian system described by coordinates x in a heat bath of inverse temperature β , we may posit a total energy E in the system+bath setup. We assume that, other than a weak thermal coupling to the bath, the system is governed by a Hamiltonian $H(x, \lambda(t))$ where λ acts as a time-varying field that we control. We can define the volume $\Omega[x(t)]$ in the microcanonical phase space of the setup as the set of states for which the system starts in $x(t)$ and subsequently traverses the trajectory $x(t)$ over time τ for a given choice of $\lambda(t)$. All such setup microstates for which the system follows $x(t)$ will gain the same value W in energy during driving, so that the setup afterwards has a total energy $E + W$. We may also define $\Omega^*[x^*(\tau - t)]$ as the volume in the phase space of the $E + W$ energy shell for which a reverse protocol $\lambda(\tau - t)$ will, when acting on the momentum reversed initial condition $x^*(\tau)$, cause the system to run along the reverse track $x^*(\tau - t)$. By time-reversal symmetry and Liouville’s theorem it must be the case that $\Omega^*[x^*(\tau - t)] = \Omega[x(t)]$.

Now, focusing on the stochastic dynamics of the system by tracing over the unknown microstate of the bath, we can say that assuming *a priori* equal likelihood of bath microstates, the probability of seeing the system traverse $x(t)$ given that it starts at $x(0)$ at $t = 0$ and is driven by $\lambda(t)$ must be given by

$$\pi[x(t)|x(0); \lambda(t)] = \frac{\Omega[x(t)]}{\Omega_{bath}[E - H(x(0), \lambda(0))]}$$

Similarly, the probability of seeing a system traverse $x^*(\tau - t)$ given it is initialized at $x^*(\tau)$ and driven by $\lambda(\tau - t)$ must be

$$\pi^*[x^*(\tau - t)|x^*(\tau); \lambda(\tau - t)] = \frac{\Omega^*[x^*(\tau - t)]}{\Omega_{bath}[E + W - H(x(\tau), \lambda(\tau))]}$$

Taking the ratio of these probabilities and exploiting the definitions of heat $Q[x(t)] = W[x(t)] - \Delta H(x, \lambda)$ and inverse temperature β for a heat bath, we obtain

$$\frac{\pi[x(t)|x(0); \lambda(t)]}{\pi^*[x^*(\tau - t)|x^*(\tau); \lambda(\tau - t)]} = \exp[\beta Q[x(t)]]$$

This is a version of the microscopic reversibility relation due to Crooks in [7].

In order to recover Eq. 1 in our text, we now simply integrate out the internal states of the ensemble of microtrajectories connecting initial to final conditions. In what follows, $\lambda(t)$ will be left implicit.

$$\begin{aligned}\pi_\tau^*[x^*(\tau) \rightarrow x^*(0)] &= \int_{(0,\tau)} \mathcal{D}x(t) \pi^*[x^*(\tau-t)|x^*(\tau)] = \\ &= \int_{(0,\tau)} \mathcal{D}x(t) \pi[x(t)|x(0)] \exp[-\beta Q[x(t)]] = \\ &= \pi_\tau[x(0) \rightarrow x(\tau)] \left(\frac{\int_{(0,\tau)} \mathcal{D}x(t) \pi[x(t)|x(0)] \exp[-\beta Q[x(t)]]}{\int_{(0,\tau)} \mathcal{D}x(t) \pi[x(t)|x(0)]} \right)\end{aligned}$$

which implies that

$$\frac{\pi_\tau^*[x^*(\tau) \rightarrow x^*(0)]}{\pi_\tau[x(0) \rightarrow x(\tau)]} = \langle e^{-\beta Q} \rangle_{x(0) \rightarrow x(\tau)}$$

From here, algebraic rearrangement and the definition of heat are sufficient to recover Eq. 1.

Moreover, in the absence of drive, the result reduces to the well-known detailed balance relation for the condition of thermal equilibrium

$$\frac{\pi_\tau^*[x^*(\tau) \rightarrow x^*(0)]}{\pi_\tau[x(0) \rightarrow x(\tau)]} = e^{-\beta \Delta E}$$

Model

The goal of this section is to describe in detail the “toy chemistry” we study in this work—a system of N particles in one dimension embedded in a heat bath at inverse temperature β with bonds (modeled as harmonic springs) between them that form and break at rates depending on inter-particle distances.

Equation of motion

Let us suppose first that the pattern of bonding—how particles are connected to one another—in our system is fixed. When bonded, particles interact as if an attractive (classical) harmonic spring of stiffness k and natural length zero connects them. When unbonded, particle pairs do not interact at all.

The Hamiltonian of our system will therefore be

$$H(x, p) = \sum_{i=0}^{N-1} \frac{p_i^2}{2m} + \frac{k}{2} \sum_{i < j}^{N-1} A_{ij} (x_i - x_j)^2 + \frac{k_0}{2} \sum_{i=0}^{N-1} x_i^2$$

where the A_{ij} are entries of the adjacency matrix defining the connectivity of our network of springs, that is, $A_{ij} = 1$ if the i th particle is bonded to the j th particle, and $A_{ij} = 0$ otherwise. There is also a weak confining harmonic potential (with constant k_0) applied to all the particles.

The presence of the thermal bath is taken to give rise to a viscous drag force on every particle, as well as a stochastic noise term (and the magnitude of these two effects is related by the fluctuation-dissipation theorem). In addition, one particle (which we always label “0”) is distinguished and is subjected to periodic forcing (of frequency ω_d and amplitude F) in one dimension. At this stage we introduce an important assumption—the noise term can be neglected. Further below, we will describe how we checked this assumption is reasonable given the parameter values we chose for our simulations.

The considerations described above yield the following equations of motion:

$$m \frac{d^2 x_i}{dt^2} + b \frac{dx_i}{dt} + k \sum_{j=0}^{N-1} A_{ij} (x_i - x_j) + k_0 x_i =$$

$$m \frac{d^2 x_i}{dt^2} + b \frac{dx_i}{dt} + k \sum_{j=0}^{N-1} L_{ij} x_j + k_0 x_i = \delta_{i0} F \sin(\omega_d t)$$

where $L_{ij} = \delta_{ij} (\sum_k A_{ik}) - A_{ij}$ (no implied summation) is the graph Laplacian (or, up to a factor of k , the “stiffness matrix”) of our spring network.

Markovian dynamics

So far, we have just described a set of N coupled damped harmonic oscillators in one dimension, subject to forcing. Now we will describe the mechanism by which bonds form and break.

When A_{ij} changes instantaneously, the positions of the particles contain information about the previous system configuration (pattern of bonding), which vanishes after a transient as the system relaxes to a new driven steady state. We make a key timescale separation assumption—that the characteristic time between bond events (formation or breakage) is much greater than the decay rate of transients in the mechanics of our spring network. This means that the memory of previous configurations is gone from the system too quickly to impact the transition rates to new configurations, and so we can view our model as a Markov process on a huge state space—all possible graphs on N nodes. In this section, we will describe how to compute the rates for this process, which form the basis of all the simulations we describe in the text, which are stochastic realizations of the process using the standard Gillespie technique [1].

The stochastic evolution of our system was mapped to a Markov process as follows:

- 1) For a given spring network, the particular solution for damped, steady-state sinusoidal driving was obtained.

- 2) This solution was used to compute the distance of separation for each pair of particles as a function of time.
- 3) This distance trajectory was used to compute an averaged Arrhenius transition rate for that bond event over the drive period.
- 4) This average rate is the Markov matrix element for that bond event.

Note that the assumption of strong timescale separation implies particles disconnected from the driven cluster relax to their mechanical equilibrium point and stay there until they re-bond to the driven cluster. Our weak confining potential causes all such particles to relax to the vicinity of the origin. In principle, Brownian fluctuations localize particles to a Gaussian about the origin of some width, but this width is very small relative to the size of the driven cluster at our chosen drive amplitude.

We begin by diagonalizing our stiffness matrix L , and writing

$$m\ddot{z}_i + b\dot{z}_i + k\lambda_i z_i = f_i \sin(\omega_d t)$$

Where the z_i are displacements of normal modes (i.e., if U is a matrix whose columns are eigenvectors of L , $x_i = U_{ij}z_j$), the λ_i are the associated eigenvalues of L , and $f_i = FU_{0i}$. It is straightforward to solve these decoupled equations for the steady state motion (“particular solution”) of each normal mode.

$$z_j(t) = \frac{f_j}{k\lambda_j - m\omega_d^2 + ib\omega_d} e^{i\omega_d t}$$

We now transform back to position space using U (note the real part Re commutes with linear operations, so we can take that at the very end), to yield $x_j(t)$. All the particles oscillate about the origin at the driving frequency, but with (in general) different amplitudes and phases. The pairwise distances between the particles will therefore also oscillate (at the driving frequency), with amplitudes given by distances between the x_j as points in the complex plane. We call the amplitude of the distance between the i th and j th particles d_{ij} .

Finally, we use these amplitudes to compute bond event rates, which follow rules derived from transition state theory with a certain barrier height (as described in the main text). The instantaneous rates for formation and breakage are:

$$r_{ij}^{form}(t) = r_0 e^{-\beta(k(x_i(t)-x_j(t))^2)}$$

$$r_{ij}^{break}(t) = r_0 e^{-\beta(\frac{1}{2}k(x_i(t)-x_j(t))^2 - \epsilon)}$$

Given our timescale separation assumption, we can average these rates over one drive cycle to find the effective rates between spring networks (the states of our Markov process).

$$\langle r_{ij}^{form} \rangle = r_0 \frac{\omega}{2\pi} \int_0^{\frac{2\pi}{\omega}} e^{-\beta(k(d_{ij}\sin(\omega t))^2)} dt = e^{-\frac{1}{2}\beta k d_{ij}^2} I_0\left(\frac{1}{2}\beta k d_{ij}^2\right)$$

$$\langle r_{ij}^{break} \rangle = r_0 \frac{\omega}{2\pi} \int_0^{\frac{2\pi}{\omega}} e^{-\beta(\frac{1}{2}k(d_{ij}\sin(\omega t))^2 - \epsilon)} dt = e^{\beta\epsilon - \frac{1}{4}\beta k d_{ij}^2} I_0\left(\frac{1}{4}\beta k d_{ij}^2\right)$$

At each step of our simulations, we need to compute the $\frac{N(N-1)}{2}$ transition rates defined by the integrals above, but we never compute the entire Markov matrix, which is prohibitively large due to the size of our state space. Any given simulation only visits a small fraction of all the possible states.

The phenomena described in the main text were observed in this model in a parameter regime where the oscillators were underdamped and in which temperature was small compared to the temperature scale $(F^2/k)/k_b$ set by k and the magnitude F of driving.

An example set of parameters satisfying these conditions, and the one used to generate the data presented in our figures, is: $\beta = 4000$, $\epsilon = 0.0001$, $b = 0.01$, $m = 1$, $k = 1$, $F = 10$. A typical number of Gillespie steps in a single stochastic trajectory included in our results was 10^4 .

We do not believe our gross results are finely sensitive to parameter choices.

Scale of motion due to thermal noise

In principle, as mentioned earlier, particles subject to viscous drag at constant temperature also experience a fluctuating random force from the thermal bath. We sought to establish that the effect of this force is negligible in our chosen parameter regime. To do so, we calculated, for typical spring networks seen in our simulations, the radius of gyration that would be observed due to thermal fluctuations alone and compared this to the radius of gyration that we would expect to observe due to forcing of a single particle at a chosen amplitude. So long as the latter is large compared with the former, we can be confident that the deterministic particular solution to the oscillator equation of motion accurately describes the typical distances between particles in the network.

The radius of gyration is defined:

$$R_g^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \langle x \rangle)^2$$

where

$$\langle x \rangle = \frac{1}{N} \sum_{i=1}^N x_i.$$

Since R_g^2 is not affected by unitary changes in coordinates, each normal mode can be taken to make a separate, summative contribution. The mean square radius of a single harmonic oscillator may easily be shown to be given $\frac{1}{m\beta\omega^2}$, and thus

$$\langle R_g^2 \rangle_\mu = \frac{1}{m\beta} \sum_{i=1}^N \frac{1}{\omega_i^2}$$

This sum is dominated by the lowest frequency normal modes (which correspond to motions that have a low stretch energy and are easily excited by thermal fluctuations).

In contrast, the radius of gyration in the presence of strong external driving is determined by the amplitude of response in the particular solution for the damped oscillator network. For the drive amplitude chosen in our simulations, the driven radius of gyration was orders of magnitude larger than in the undriven case across the whole range of relevant drive frequencies, providing strong support for the choice to ignore the diffusive effect of thermal Brownian noise on the motions of individual particles.

To confirm this, we ran full molecular dynamics simulations with time dependent rates and noise. Figure S1 shows the contributions to the radius of gyration from thermal noise (which we neglect) and from the deterministic parts of the dynamics (that contribute to the rates of our Markov process on graphs described above).

The arguments in this section do not hold when dealing with the equilibrium ensemble—the absence of the drive in that case means there is no natural length scale relative to which motion due to noise is small. However, the spectrum of natural frequencies in the equilibrium ensemble computed from the simulation including noise agrees closely with the same computed from Gillespie realizations of our system neglecting noise and with rates averaged over a drive cycle.

Results

Characteristics of typical networks in the driven ensemble

It is natural to ask how structural characteristics of the typical networks drawn from the driven ensemble differ from those of typical networks seen in the undriven case.

In each simulation, we equilibrated the system and then turned on the drive and propagated the Markov dynamics. We found that in general, the number of bonds (edges) in the network tended to relax to a different value in the driven steady state than at equilibrium. (Figures S2 and S3). However, other than being somewhat more dispersed, the degree (edges per node) distribution of the driven ensemble did not show significant qualitative differences compared to the equilibrium ensemble (Figure S4).

As a more sophisticated measure of network structure, we looked at the following global property of the adjacency matrix A_{ij} . The trace of a power A^m of the adjacency matrix A gives the number of m step walks on the graph that return to their starting point. The largest eigenvalue of A , denoted here as ϵ_{max} , therefore gives the growth rate for the number of these walks, which in turn quantifies the overall connectedness of the network. With this metric we can compare the whole ensemble of both driven and undriven systems and examine the distributions, which are presented in Figures S5 and S6. The driven ensemble distribution is shifted towards less positive values of ϵ_{max} relative to the

undriven distribution, implying a lesser interconnectedness that reflects system-wide topological correlations that are atypical in the equilibrium ensemble.

Entropy production

In the main text, we refer several times to “stochastic entropy production”. The way we calculated this is by using the coarse-grained fluctuation relation [2] which sets a lower bound on the total entropy change associated with a transition between A and B:

$$\Delta S_{\min} \geq \ln \frac{\pi(A \rightarrow B)}{\pi(B \rightarrow A)}$$

where $\pi(A \rightarrow B)$ and $\pi(B \rightarrow A)$ are forward and reverse transition rates between states. In our work, the states are spring networks and the transition rates are those we describe how to compute in the previous section. For example, for the event of a bond forming between particles i and j , we have:

$$\Delta S_{\min} \geq \ln \frac{r_{ij}^{\text{form}}(t_n)}{r_{ij}^{\text{break}}(t_{n+1})}$$

where t_n, t_{n+1} are consecutive Gillespie times.

Average and transient work

In a driven steady state of our spring system, the average power dissipated (over a drive cycle) is:

$$\langle P \rangle = \frac{b\omega_d^2}{2} \sum_i |x_i|^2$$

where $|x_i|$ is the amplitude of the periodic motion of the i th particle.

Using the expression for $\langle P \rangle$ we can now calculate the transient work ΔW . We define the transient work to be the additional work absorbed from the drive during a transition in excess of the work that would be absorbed if the instantaneous rate of dissipation were always equal to the average dissipation rate for the given spring network. This quantity is of interest because it measures the portion of the work absorbed during a transition between steady states that can contribute to irreversibility, as opposed to that dissipated in the driven steady states achieved before and after the transition, which cannot [3, 4].

In order to calculate the transient work, we first calculate the dynamics of the particles under a certain spring network topology, which is, as usual, that of a simple damped harmonic oscillator. Let us denote the dynamics in this interval as I. We use these dynamics to calculate the time-dependent rates, and note the most probable time T for a bond event during a drive cycle. We then suppose the bond event occurs, and the network structure is updated accordingly. The dynamics of the new structure, labeled II, are then

computed. See Figure S7 for a schematic of this procedure. For each segment of dynamics we then calculate the transient work by taking the integrated work up to/from the most probable time of bond event $\mathcal{T}$.

$$\Delta W_I = \int_{-\infty}^{\mathcal{T}} (\dot{x}_0 F_{drive} - \langle P \rangle_I) dt$$

$$\Delta W_{II} = \int_{\mathcal{T}}^{\infty} (\dot{x}_0 F_{drive} - \langle P \rangle_{II}) dt$$

where $\langle P \rangle_I$ and $\langle P \rangle_{II}$ are the average power $\langle P \rangle$ dissipated (over a drive cycle) for the structures I and II, respectively. The total transient work is then simply the sum

$$\Delta W = \Delta W_I + \Delta W_{II}$$

An underlying assumption made here to simplify calculations is that the bond event of interest occurs at the time it is *most likely* to occur. We performed a simple check to verify this was good assumption by taking the time evolution of the distance between two particles over a drive cycle and using it to compute the time evolution of the rate for the associated bond event (data not shown). We found in the typical case that the rate was sharply peaked at one time, and dropped to less than half its value in a time small compared with a single drive period, thus justifying our approximation.

Bonds that snap

In the bulk of this work, the rate equations obeyed by the bonds can properly be thought of as modeling “catch bonds” because their rate of breakage decreases as they stretch. We have also investigated a different model which still makes bond *formation* likely at short distances, just as in the case of catch bonds. However, in this “snapping” model, bonds are likeliest to break at *large* distances of particle separation, i.e. their rate of breakage increases when they are stretched. This might correspond, intuitively, to interactions such as van der Waals attraction that weaken at long range.

To model this kind of bond, we used a barrier energy function that includes soft repulsion in addition to a harmonic pairwise interaction:

$$B(x) = \frac{1}{2} k x^2 + k e^{-|x|}$$

The resulting rate integrals:

$$\langle r_{ij}^{form} \rangle = r_0 \frac{\omega}{2\pi} \int_0^{\frac{2\pi}{\omega}} e^{-\beta \left(\frac{k}{2} (d_{ij} \sin(\omega t))^2 + k e^{-|d_{ij} \sin(\omega t)|} \right)} dt$$

$$\langle r_{ij}^{break} \rangle = r_0 \frac{\omega}{2\pi} \int_0^{\frac{2\pi}{\omega}} e^{-\beta \left(k e^{-|d_{ij} \sin(\omega t)|} - \epsilon \right)} dt$$

appear to have to have no simple closed form solution, and were solved numerically.

We chose to pick parameters in the snapping case so that typical rates were of similar orders of magnitude to those in our other simulations, but our qualitative results appear to not be finely sensitive to these choices. A precise matching between the parameter

choices in the “catch” and “snap” rules is difficult because we did not non-dimensionalize our model. As a consequence of this, for example, there is a length scale in the argument of the exponential in the barrier height that we have arbitrarily set to 1.

Simulations of our model using these different rate rules gave, broadly, results reversed relative to those reported in the main text. Strikingly, the spectrum of such systems showed a suppression effect in frequencies around the driving frequency. An example of this effect is illustrated in Figure S8. Furthermore, the projection of the force vector in the normal mode basis shows that the same frequencies also tend to decouple from the drive.

Supplementary References

1. Cao, Y., Gillespie, D. & Petzold, L. Multiscale stochastic simulation algorithm with stochastic partial equilibrium assumption for chemically reacting systems. *Journal of Computational Physics* 206, 395-411, (2005).
2. England, J. Statistical physics of self-replication. *Journal of Chemical Physics* 139, 121923 (2013).
3. Marsland, R. & England, J. Far-from-equilibrium distribution from near-steady-state work fluctuations. *Physical Review E* 92, (2015).
4. Perunov, N., Marsland, R. M. & England, J. L. Statistical physics of adaptation. *Physical Review X* 6, 021036 (2016).

Supplementary Figures

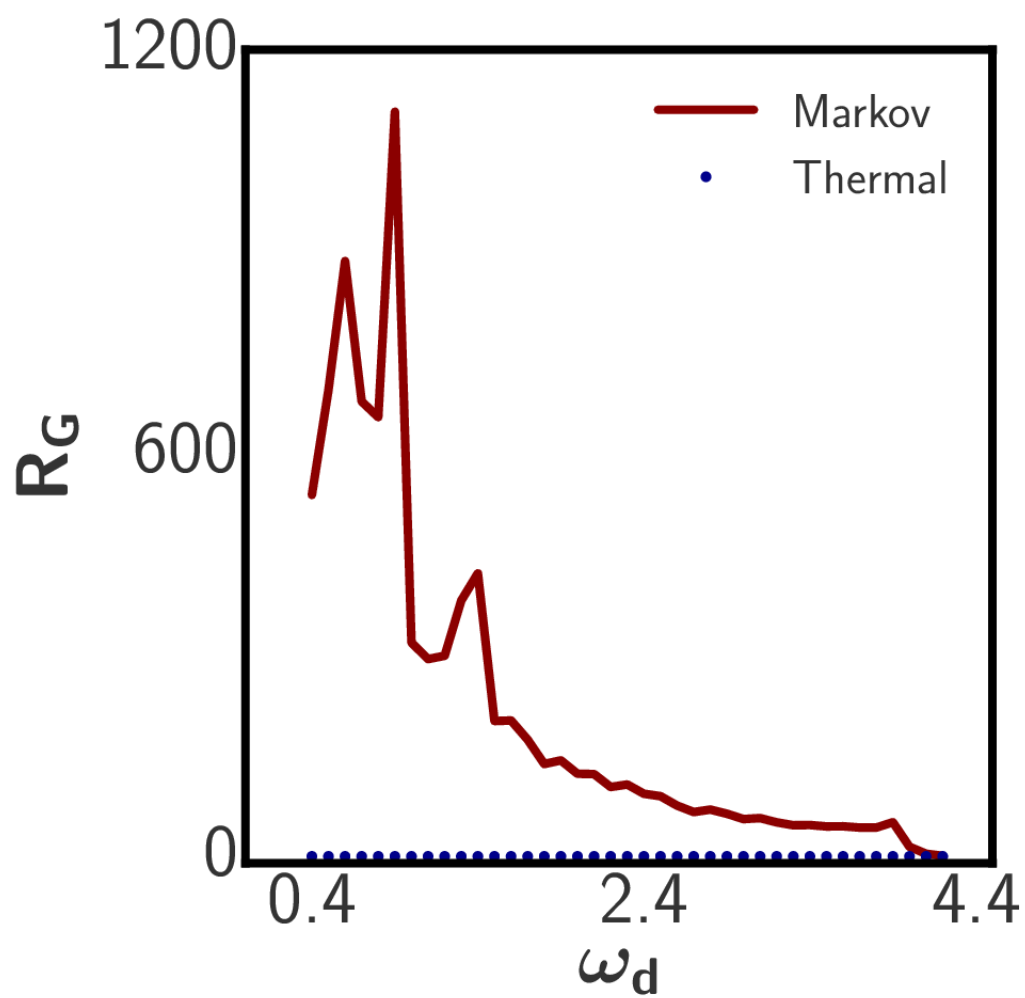


Figure S1. Contributions to the radius of gyration from thermal noise (blue) and from the deterministic parts of the dynamics (red) over a range of values of driving frequency.

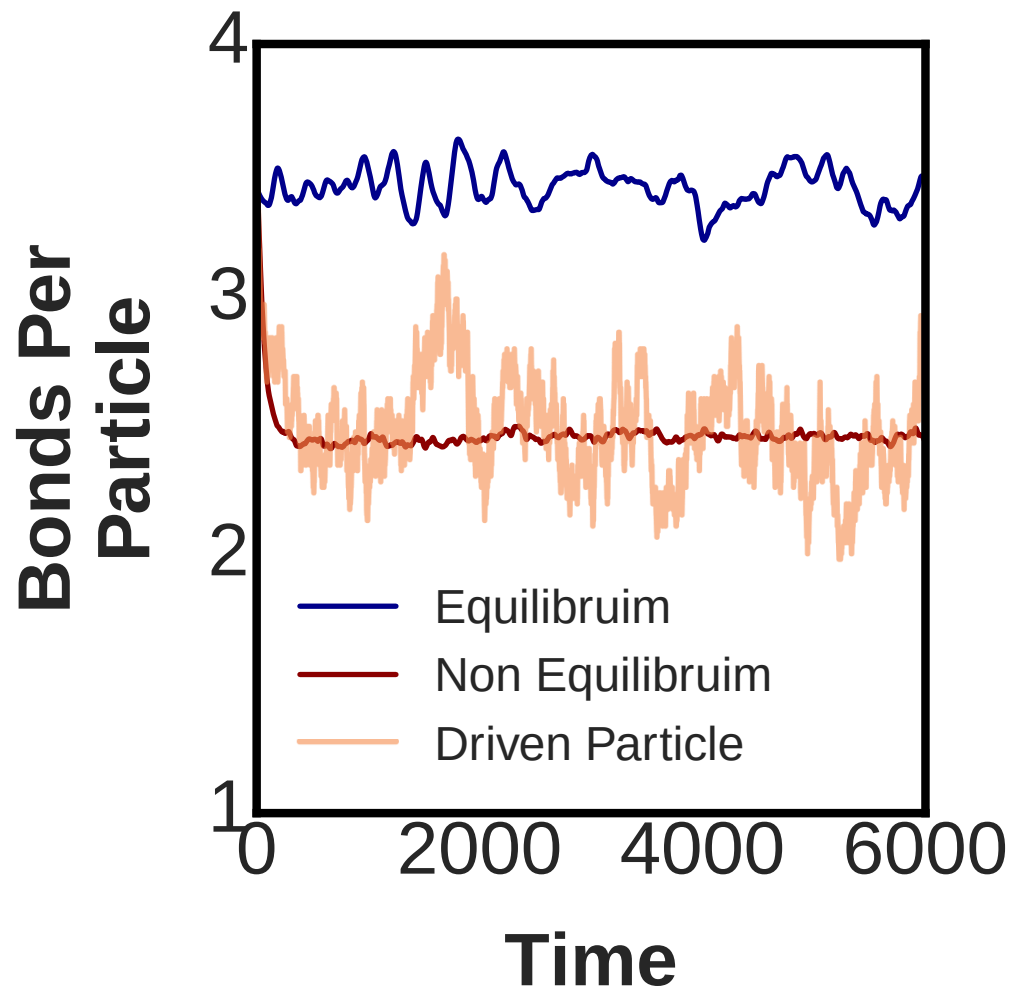


Figure S2. Red: the evolution over time of the average number of bonds per particle as the system is driven away from equilibrium with $\omega_d = 1.3$. Orange: the number of bonds attached to the driven particle over the course of the same system trajectory. Blue: an undriven propagation of the same initial conditions.

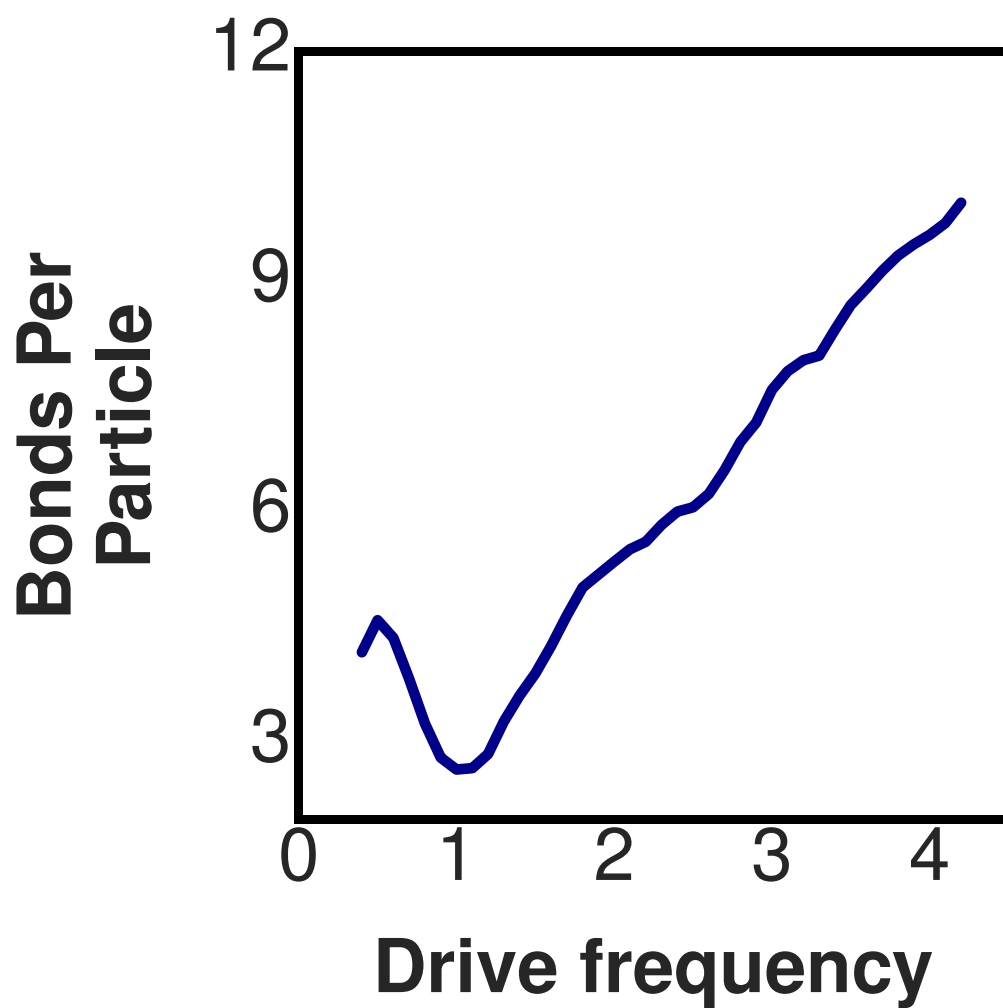


Figure S3. The average number of bonds per particle in the nonequilibrium steady state, as a function of drive frequency.

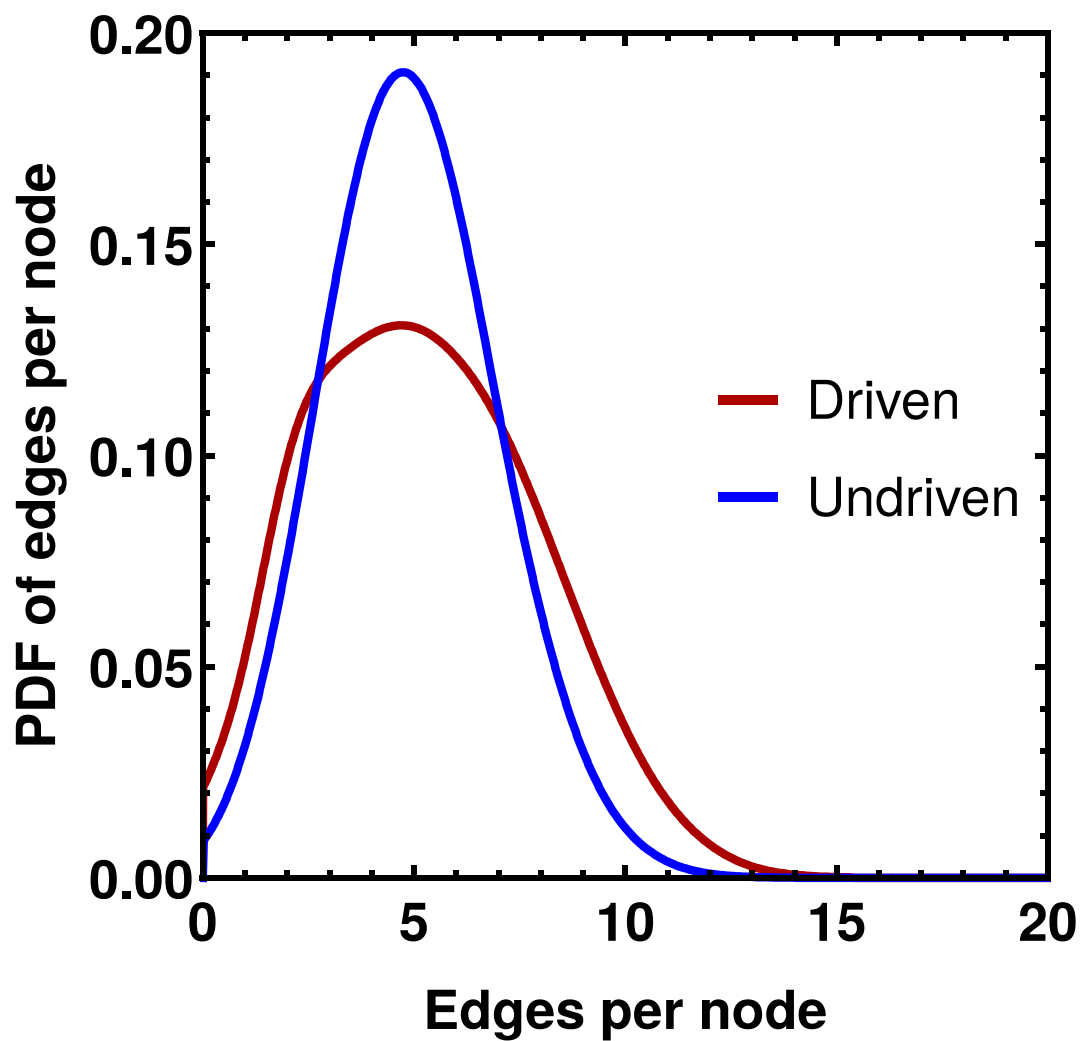


Figure S4. The degree distribution in a driven ($\omega_d = 1.2$) and undriven ensemble of spring networks.

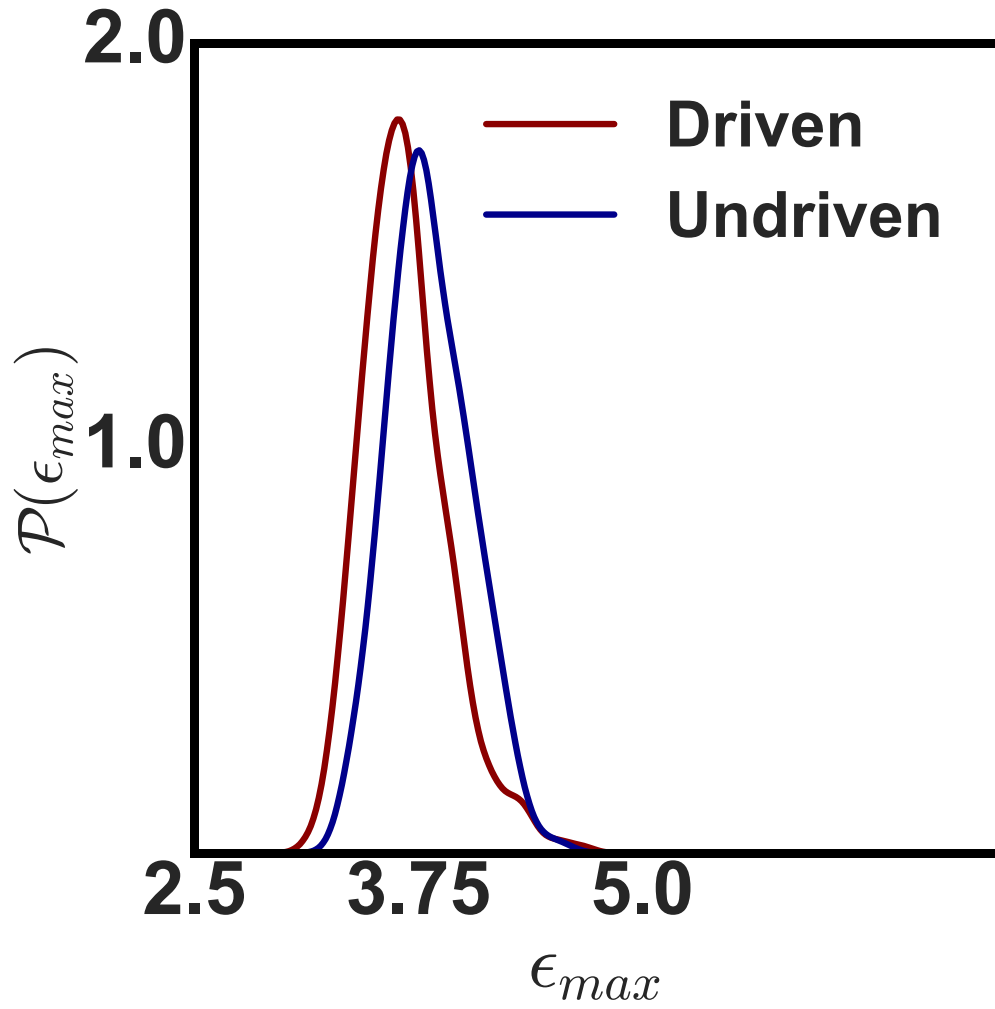


Figure S5. Distribution of the largest eigenvalue of the adjacency matrix of graphs drawn from a nonequilibrium steady state ($\omega_d = 1.2$), whose number of bonds is equal to the steady state average.

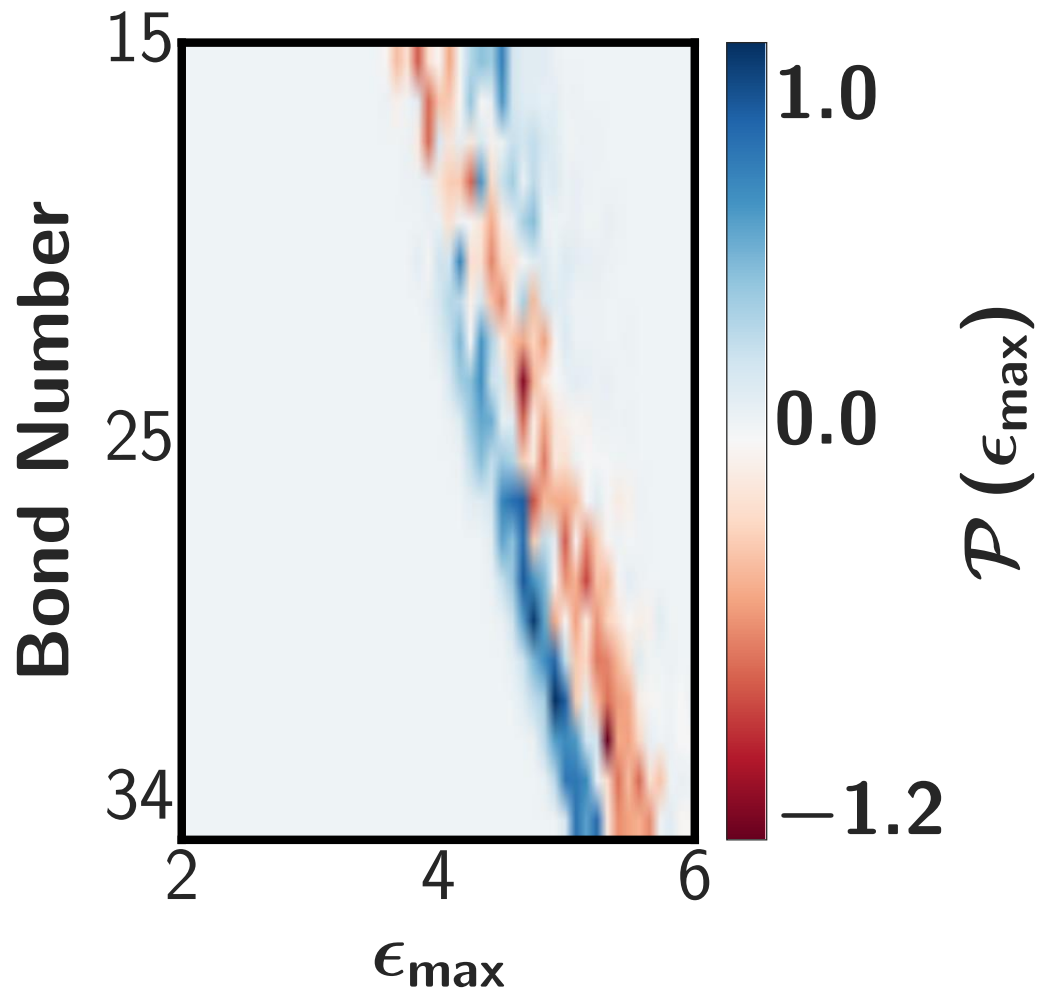


Figure S6. The probability density function of the largest eigenvalue of the adjacency matrix for ensembles of graphs with a range of fixed bond numbers, in the driven case, *minus* the same in the undriven case. Each horizontal slice of this figure corresponds to plotting the difference of the two curves in Figure S5, for some fixed bond number (Figure S5 is the case where that fixed number is the steady state average).

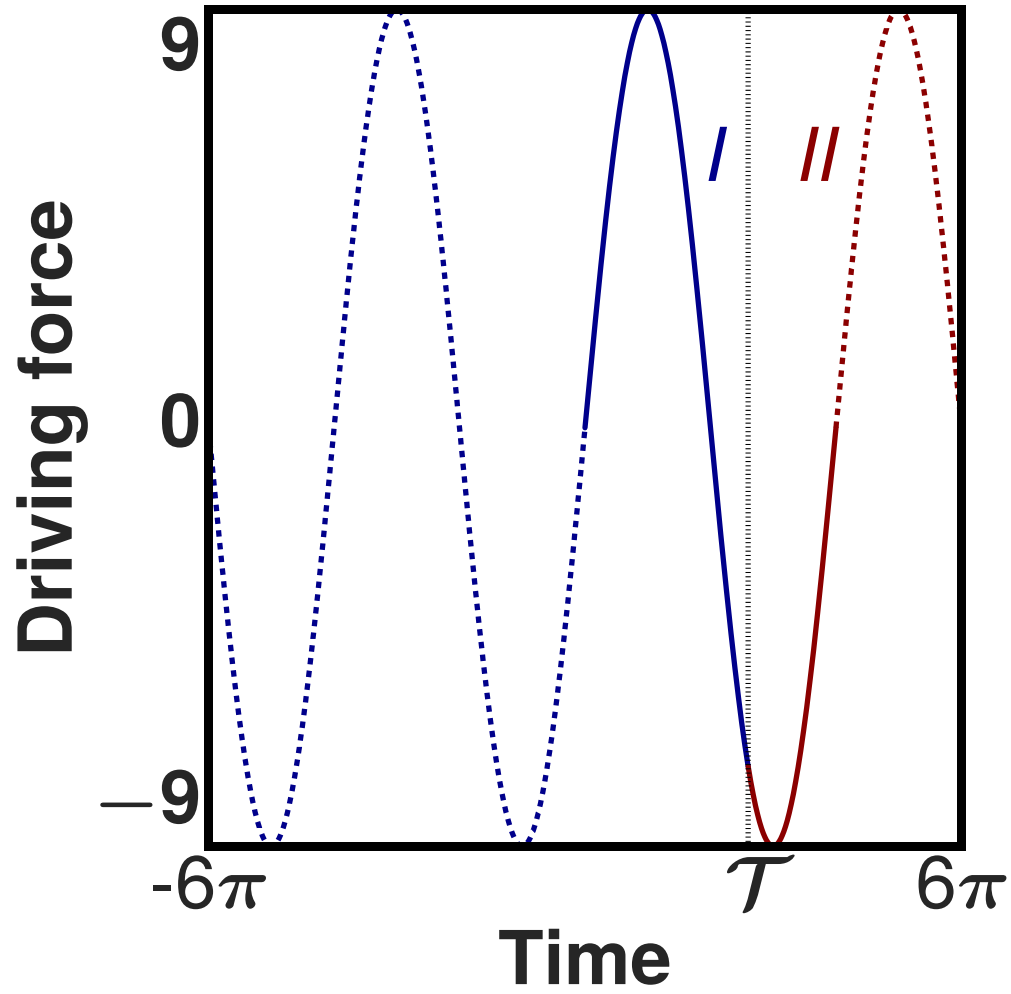


Figure S7. A bond forms or breaks at some point T of the drive cycle, changing a spring network topology, labeled I, to a new one, labeled II. When this happens, excess work is done over what would be dissipated in steady state, and this has contributions from the incomplete drive cycle (blue solid line) and from the relaxation to the new steady state after T .

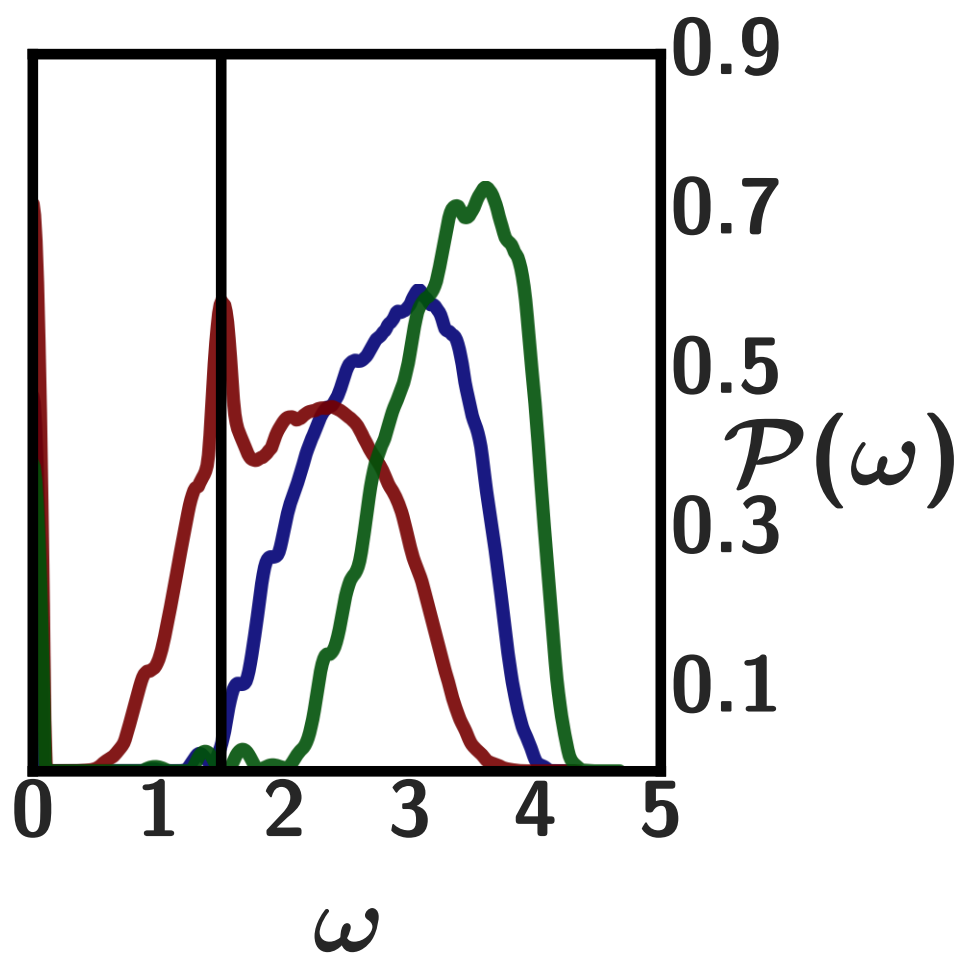


Figure S8. Distribution of natural frequencies for the driven system ($\omega_d = 1.5$) with bonds that “catch” (red), compared to the same with bonds that “snap” (green) and the undriven system (blue).

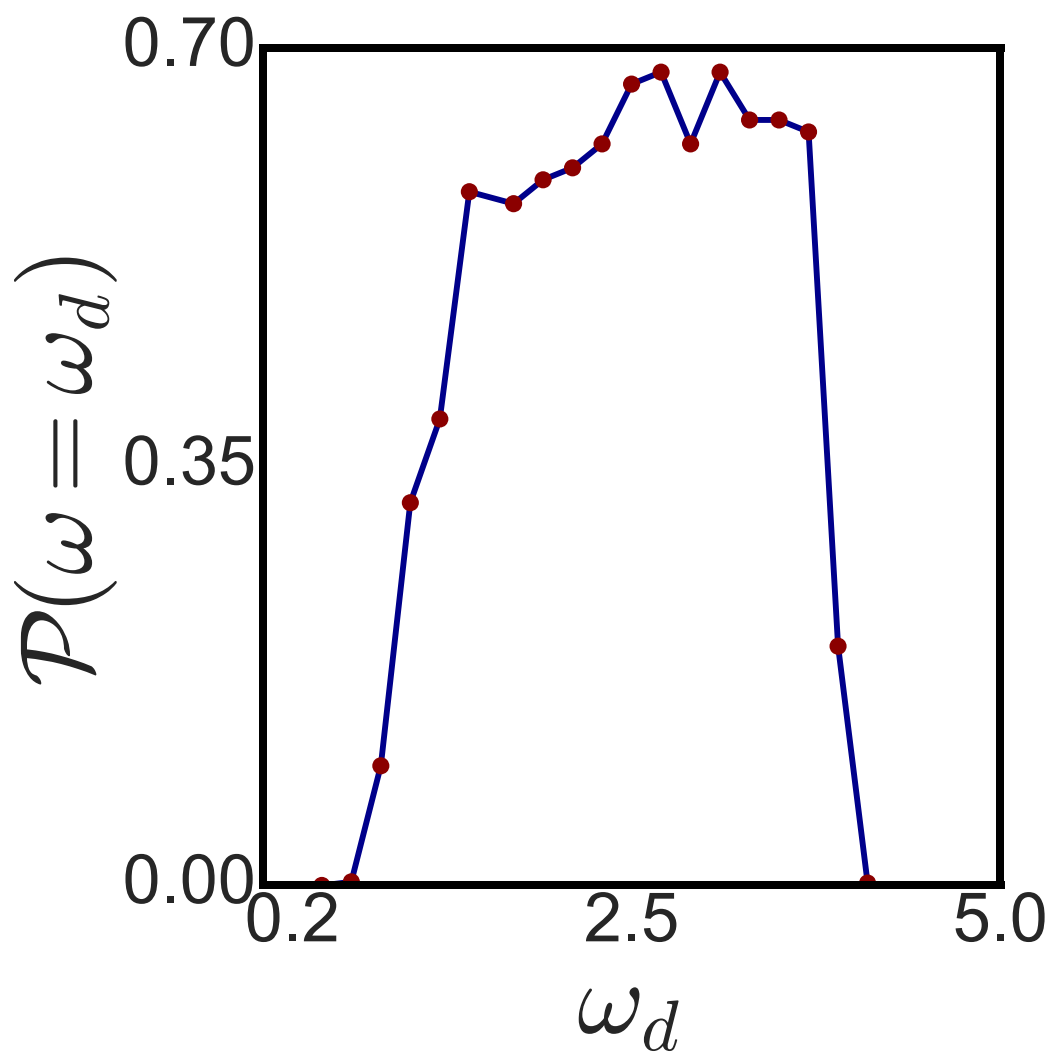


Figure S9. The height of the normal mode distribution evaluated at the driving frequency is plotted as a function of driving frequency. The value loosely follows the shape of the equilibrium “haystack” normal mode spectrum.